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1. The matrix M represents the transformation T and is given by

$$M = \begin{pmatrix} p+3 & -2 \\ p+2 & 1-p \end{pmatrix}$$

- (a) In the case when $p = -1$ show that the image of the point $(3, 4)$ under T is the point $(-2, 11)$. [2]
- (b) In the case when $p = 2$ find the gradient of the line of invariant points under T . [3]

2. A linear transformation of the plane is represented by

$$\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$$

Determine the invariant line with negative gradient.

[4]

3. The matrix A is given by

$$A = \frac{1}{5} \begin{pmatrix} -3 & -4 \\ -4 & 3 \end{pmatrix}$$

You are given that A represents the transformation T , where T is a reflection in a straight line through the origin, O .

- (a) Explain why T must have a line of invariant points. State the geometric significance of this line. [2]
- (b) By considering the line of invariant points for T , determine the equation of the mirror line.
Give your answer in the form $y = mx + c$. [4]
- (c) The coordinates of the point P are $(4, 7)$.
By considering the image of P under the transformation T , or otherwise, determine the coordinates of the point on the mirror line which is closest to P . [3]
- (d) The line with equation $2x + by = 6$ is an invariant line for T .
Determine the value of b . [2]

4. The matrix M is given by

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 6 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

- (a) The matrix M represents a sequence of two geometrical transformations in the x - y plane.
Give full details of the two transformations, stating clearly the order in which they are applied. [4]
- (b) Write M^{-1} as the product of two matrices, neither of which is I . [2]
- (c) Find the equations of the invariant lines, through the origin, of the transformation represented by M . [5]

5. A linear transformation of the plane is represented by the matrix

$$\mathbf{M} = \begin{pmatrix} 2 & -1 \\ p & 0 \end{pmatrix}$$

where p is a constant.

- (a) Show that the line $x = 0$ is not invariant. For a line through the origin with equation $y = mx$, show that it is invariant only if

$$m^2 - 2m + p = 0$$

Hence find the set of values of p for which the transformation has no invariant lines through the origin. [5]

- (b) Given that the transformation multiplies areas by 3 and reverses orientation, find the invariant lines. [3]

6. The 2×2 matrix $\mathbf{A}$ represents a transformation $\mathbf{T}$ which has the following properties.

- The image of the point $(1, -1)$ is the point $(2, 0)$.
- An object shape whose area is 4 is transformed to an image shape whose area is 24.
- $\mathbf{T}$ has a line of invariant points.

(i) Find a possible matrix for $\mathbf{A}$.

[8]

The transformation $\mathbf{S}$ is represented by the matrix $\mathbf{B}$ where

$$\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$$

(ii) Find the equation of the line of invariant points of $\mathbf{S}$.

[2]

(iii) Show that any line of the form $y = 2x + c$ is an invariant line of $\mathbf{S}$.

[3]

7. The matrix

$$B = \begin{pmatrix} 1 & -4 \\ 2 & -1 \end{pmatrix}$$

represents the linear transformation T .

Any invariant line for T must be parallel to an invariant line through the origin.

- (a) A line through the origin has equation $y = mx$, where m is real. Show that, for $m \neq \frac{1}{4}$, its image under T has equation

$$y = \frac{2-m}{1-4m}x$$

State also the image of the line $y = \frac{1}{4}x$ and the image of the vertical line $x = 0$.

[3]

- (b) Hence prove that T has no invariant lines.

[2]

8. The matrix of a linear transformation T is

$$M = \begin{pmatrix} p+2 & -5 \\ -2 & p-1 \end{pmatrix}$$

(a) Find the values of p for which T is not invertible.

[2]

It is now given that $p = 3$, so that

$$M = \begin{pmatrix} 5 & -5 \\ -2 & 2 \end{pmatrix}$$

(b) Find the equations of the invariant lines, through the origin, of the transformation represented by M .

[6]

9.

$$A = \begin{pmatrix} b & a-3 \\ 2a-1 & -2 \end{pmatrix}$$

where a and b are non-zero constants.

Given that the matrix A is self-inverse,

- (a) determine the value of b and the possible values of a . [5]

The matrix A represents a linear transformation M of the plane.

Using the smaller value of a from part (a),

- (b) show that the invariant points of the linear transformation M form a line, stating the equation of this line. [3]